1.1 BIDMAS

When carrying out calculations involving more than one mathematical operation, it is important to follow a standard order. This ensures that everyone obtains the same answer and avoids ambiguity.

For example, consider the calculation

$$3 + 4 \times 5$$

Should the addition be performed first or the multiplication? A standard convention is therefore required.

Definition

BIDMAS is a rule used to determine the order in which mathematical operations are carried out.

B	Brackets
I	Indices
D	Division
M	Multiplication
A	Addition
S	Subtraction

Operations higher in the list are performed before those lower down.

Division and multiplication have equal priority and are evaluated from left to right.

Addition and subtraction likewise have equal priority and are evaluated from left to right.

1.1.1 Using BIDMAS

Consider the following:

Worked Example

Evaluate

$$3 + 4 \times 5$$

The multiplication must be performed first:

Substituting this result gives

Therefore,

If the addition was performed first, the answer would be

$$3 + 4 \times 5 = 35$$

which is incorrect because it does not follow BIDMAS.

Engineering Application

Engineers frequently perform calculations involving several mathematical operations.

Consider an engineering workshop producing components. A machine setup requires a fixed 10 minutes before production can begin and each of the 4 components takes 5 minutes to manufacture.

The total production time is

$$\begin{aligned} 10 + 5 \times 4 &= 10 + 20 \\ &= 30 \text{ minutes} \end{aligned}$$

It would not make physical sense to calculate left to right

$$\begin{aligned} 10 + 5 \times 4 &= 15 \times 4 \\ &= 60 \text{ minutes} \end{aligned}$$

because the setup time is incurred only once, whereas the manufacturing time is repeated for each component.

Applying the correct order of operations ensures that calculations are consistent, reproducible and safe. A simple mistake in the order of operations can lead to significant errors.

1.1.2 Brackets

Anything inside brackets must be evaluated first.

Worked Example

Evaluate

$$(9 + 2) \times 3$$

First calculate the expression inside the brackets:

Then multiply:

Therefore,

Notice that brackets completely change the answer.

1.1.3 Indices

Indices (powers) are performed after brackets and before multiplication.

Definition

An index tells us how many times a number is multiplied by itself. For example,

$$4^2 = 4 \times 4 = 16$$

and

$$2^3 = 2 \times 2 \times 2 = 8$$

Worked Example

Evaluate

$$5 + 2^3$$

Calculate the index first:

Then add:

Therefore,

1.1.4 Working from Left to Right

A common misunderstanding is that division must always be carried out before multiplication because the letter D appears before M in BIDMAS. This is not true. Multiplication and division have equal priority. When both operations appear in the same calculation, they must be performed from left to right.

Worked Example

Evaluate

$$24 \div 6 \times 2$$

Since division and multiplication have equal priority, work from left to right.

Firstly,

Then,

Therefore,

Notice that it would be incorrect to calculate

$$6 \times 2 = 12$$

first and then obtain

$$24 \div 12 = 2$$

Multiplication does not automatically take priority over division.

Similarly, addition and subtraction have equal priority and are evaluated from left to right.

Worked Example

Evaluate

$$15 - 6 + 2$$

Working from left to right,

and

Therefore,

1.1.5 Common Mistakes

Common BIDMAS mistakes include:

- Ignoring brackets.
- Adding before multiplying.
- Performing multiplication before division regardless of position.
- Forgetting to evaluate indices.
- Reading expressions from right to left.

Using the fraction button on your calculator generally prevents BIDMAS errors.

1.1.6 Exercises

Exercise

Evaluate the following expressions using BIDMAS.

1. $4 + 3 \times 6$
2. $(4 + 3) \times 6$
3. $20 - 2^3$
4. $24 \div 6 \times 3$
5. $15 - 5 + 7$
6. $2(3 + 5)$
7. $18 \div (3 + 3)$
8. $5 + 4^2$
9. $7 + 2 \times (8 - 3)$
10. $(12 - 4)^2$

Answers to Exercise 1.1.6

1. 22
2. 42
3. 12
4. 12
5. 17
6. 16
7. 3
8. 21
9. 17
10. 64